

an entry is made in the database. If, and when, the stolen vehicle passes through toll plaza, the ID of the tag fixed on it matches with the stolen vehicle's tag ID in the database of the server and the vehicle gets identified as stolen.

All toll plazas must be connected to each other with a centralised server in the form of LAN. Updates of any sort of transaction are immediately updated in the server. Any vehicle violating traffic signals is detected by RFID readers fixed at signal crossings, and the traffic police get notified. Also, any vehicle travelling beyond the legal speed limit can be tracked.

Electronic toll collection system based on GPS

GPS-based system is an innovative combination of satellite-based GPS and Global System for Mobile telecommunication (GSM) technologies. GPS technology is used to find the position of a vehicle. It works in almost any condition and is accurate within 3 to 15 metres.

GPS-based electronic toll collection system consists of automatic toll collection system (ATCS), RF transmitter, toll gate and an on-board unit (OBU) mounted in the vehicle. The OBU includes an MCU, a GPS receiver and a GSM unit. It automatically determines the distance driven on the toll route by using GPS satellite signals and other positioning sensors. It then calculates the toll to be paid and transmits this information to the ATCS for further processing.

The OBU is supported with required software, electronic road maps, user data and data related to toll charges for the roads. When the vehicle approaches the toll road, information is transmitted to the Android device through GSM module, and a voice call and/or an alert SMS is sent to the user.

The on-board LCD displays vehicle details and other selected texts. An alert message or voice output enables the user to decide whether to quit the toll road (if alternate toll-free road is available) or continue using it. If the user decides to continue, a pre-determined toll amount is deducted auto-



Fig. 7: Smartphone being used to pay highway toll via an online payment service app (Credit: <https://theaseanpost.com>)

matically from the account on exiting the toll gate. Toll amount charged is based on the vehicle's emission category, number of axles and length of the toll route.

Vehicle position data is approved by GPS system and delivered to ATCS by GSM network. The MCU takes input location of the vehicle from GPS, compares it with the position of the toll plaza and updates vehicle database in ATCS. Once the system verifies that the vehicle is registered in the database, the RF transmitter sends the signal to the toll unit to open the toll gate. Thereafter, toll charge is debited from the account of the user and it is acknowledged with an SMS to the customer. If there is no balance in the account of the user, the bill is sent to the user with a fine. If the vehicle is found not registered in the database, it is diverted to the manual toll collection booth.

International technical specification

Technical specification ISO/TS 12813:2009 Electronic Fee Collection - Compliance Check Communication For Autonomous Systems was recently issued by International Organization for Standards (ISO). It lays down the specifications and requirements that an electronic toll collection system must confirm to, to be certified as a world-class system.

The standard helps in harmonising various systems to facilitate mobility between different road networks and to ensure reliable data collection and proper charging. It covers the interface between the OBU and roadside equipment, information data flow between operators, and

electronic toll collection systems based on DSRC, GSM, Global Navigation Satellite Systems (GNSS) and integrated circuit (IC) card technologies. The standard includes not only the requirements but also the associated test procedures to support conformity evaluation of the systems. It also includes security guidelines that can be useful in the preparation or evaluation of security requirements.

End note

Today, toll-related technologies can leverage mobile telecommunication networks that make operational processes such as online registration and account topup a lot simpler. However, capital expenditure (CAPEX), that is, preliminary cost of technology and establishment of the system and operational expenditure (OPEX), that is, expense of upholding online connection, Multimedia Messaging Service (MMS) and customer support costs are likely to become considerable.

Many different technologies are competing. For example, ANPR method, which is also known as video tolling, is an evolved technology and has already been deployed to some extent in Europe.

Smartphones and other mobile communications hold a great promise in terms of being deployed, given the high penetration among users. However, there are hurdles such as ascertaining that smartphones should exchange data with vehicles to ensure correct functioning in secure mode, and that these should be plugged to an energy source. This would tantamount to an intricate integration of various products that have diverse life cycles.

Also, as autonomous and connected vehicles materialise commercially, it is probable that automobiles will be ready with added short-range communication devices that use vehicle-to-everything (V2X) communication protocols [include vehicle-to-vehicle (V2V) and vehicle-to infrastructure (V2I) communications] in the 5.9GHz frequency band. Perhaps, this could be facilitated to sustain tolling in the future. **EFY**